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Project: Data-Driven Segmentation for Chestnut Ridge

Contents

List of Tables

List of Figures

Introduction

Customer-centricity has emerged as a critical strategic necessity in the evolving retail landscape (Manoj Kumar Dash et al., 2023). From offering apparel to home goods and expanding across multiple sales channels such as brick-and-mortar, catalogue and e-commerce, Chestnut Ridge is a heritage retail brand known for its diverse offerings. The company faces increasing complexity in understanding its customer base due to diversity in products and channels. According to the company's Chief Marketing Officer (CMO), a one-size-fits-all marketing strategy no longer suffices. This report aims to address the pressing need to uncover distinct customer segments and design targeted marketing strategies accordingly.

The report begins with segmentation using both hierarchical and k-means clustering on a customer survey dataset with 200 observations. In the dataset, each respondent has rated the importance of store attributes such as low prices, return policy, service quality, variety of choice electronics and furniture on a 1–10 scale. The income and age of the respondents are also included in the dataset as demographic variables to enable a robust customer segmentation analysis. This is followed by detailed segment profiling to create 4 segments and a strategic evaluation of each segment using the GE matrix. Finally, the report recommends strategies to target each segment and the most attractive segment for Chestnut Ridge based on its internal capabilities.

Section 1: Data Analysis and Customer Segmentation

Task 1: Data Inspection and Descriptive Statistics

Code:

```
# Task 1:


library(tidyverse)
```

```
retailer <- read_csv("retailer.csv")

glimpse(retailer)

summary(retailer)
```

Useful Output:

Rows: 200

Columns: 9

Table 1 Descriptive Statistics of Unnormalized data

Variable	Min	1st Qu.	Median	Mean	3rd Qu.	Max
respondent_id	1	50.75	100.5	100.5	150.25	200
variety_of_choice	4	6	8	7.57	10	10
electronics	1	3	4.5	4.45	6	10
furniture	0	1	2	3.27	6	7
quality_of_service	1	2	3	3.53	4	9
low_prices	1	2	5	4.8	7	10
return_policy	1	3	4	4.25	6	10
income	13	15	19.5	32.17	54.25	95
age	21	24	27	32.52	38	68

Discussion:

- Variables differ significantly in scale. income ranges from 13 to 95, while furniture scores range from 0 to 7. This highlights scale inconsistency.
- This inconsistency necessitates z-score normalization to ensure that clustering is not biased by high-variance variables like income and age.

- The summary() function acts as a diagnostic tool. It reveals central tendency and dispersion.
- tidyverse was used for data wrangling and visualization.

Task 2: Identification of Minimum and Maximum Values in Normalized Data

Code:

```
# Task 2:

retailer_norm <- scale(retailer[, -1]) %>% as_tibble()

mean(retailer_norm$income)

sd(retailer_norm$low_prices)

summary(retailer_norm)
```

Useful Output:

mean(retailer_norm$income) = -6.359008e-17

sd(retailer_norm$low_prices)= 1

Table 2 Descriptive Statistics of Normalized data

Variable	Min	1st Qu.	Median	Mean	3rd Qu.	Max
variety_of_choice	-1.77	-0.78	0.22	0	1.21	1.21
electronics	-1.78	-0.75	0.03	0	0.80	2.86
furniture	-1.38	-0.96	-0.54	0	1.15	1.57
quality_of_service	-1.1	-0.67	-0.23	0	0.21	2.36
low_prices	-1.50	-1.10	0.08	0	0.87	2.06

return_policy	-1.59	-0.61	-0.12	0	0.85	2.81
income	-0.84	-0.76	-0.56	0	0.97	2.77
age	-0.97	-0.72	-0.46	0	0.46	2.99

Discussion:

- electronics exhibits the lowest normalized minimum value of –1.78. This indicates that certain respondents rated its importance far below the average.
- age shows the highest standardized maximum value of 2.99. This highlights significant age variation across the sample.
- All variables were successfully standardized (mean ≈ 0, sd = 1). This ensures fair weight distribution in clustering.
- The extreme values in electronics and age highlight their discriminatory influence in defining heterogeneous customer segments.
- The summary() function is again used as a diagnostic tool.

Task 3: Data Preparation and Z-score Normalization

Code:

```
# Task 3:

store_attributes <- retailer %>%

select(variety_of_choice, electronics, furniture, quality_of_service, low_prices, return_policy)

store_attributes_norm <- scale(store_attributes) %>% as_tibble()

mean(store_attributes_norm$low_prices)
```

sd(store_attributes_norm$quality_of_service)

summary(store_attributes_norm)

Useful Output:

mean(store_attributes_norm$low_prices) = 4.266064e-17

sd(store_attributes_norm$quality_of_service) = 1

Table 3 Descriptive Statistics of Normalized store attributes

Variable	Min	1st Qu.	Median	Mean	3rd Qu.	Max
variety_of_choice	-1.77	-0.78	0.22	0	1.21	1.21
electronics	-1.78	-0.75	0.03	0	0.80	2.86
furniture	-1.38	-0.96	-0.54	0	1.15	1.57
quality_of_service	-1.1	-0.67	-0.23	0	0.21	2.36
low_prices	-1.50	-1.10	0.08	0	0.87	2.06
return_policy	-1.59	-0.61	-0.12	0	0.85	2.81

Discussion:

- The six store attributes were successfully extracted and z-score standardized (mean ≈ 0, sd = 1).
- electronics displays the highest standardized maximum value of 2.86. This indicates that some respondents place exceptionally high importance on this attribute. The lowest minimum again appears in electronics of –1.78.
- This makes electronics as the most polarized attribute in customer perceptions.
- return_policy also shows high variability (max value: 2.81, min value: –1.59). This confirms its strategic importance to specific customer cohorts.

Task 4-7: Euclidean Distance, Seed, Ward.D2, Dendrogram

Code:

```
# Task 4:

distances <- dist(store_attributes_norm, method = "euclidean")

as.matrix(distances)[1:5, 1:5]

# Task 5:

set.seed(123)

# Task 6:

hc <- hclust(distances, method = "ward.D2")

# Task 7:

plot(hc)
```

Useful Output:

Table 4 Euclidean Distance of first 5 respondents

	1	**2**	**3**	**4**	**5**
1	0.00	3.12	4.07	3.70	3.20
2	3.12	0.00	2.80	3.23	3.43
3	4.07	2.80	0.00	1.62	4.00
4	3.65	3.23	1.62	0.00	3.60

5	3.20	3.43	4.00	3.60	0.00

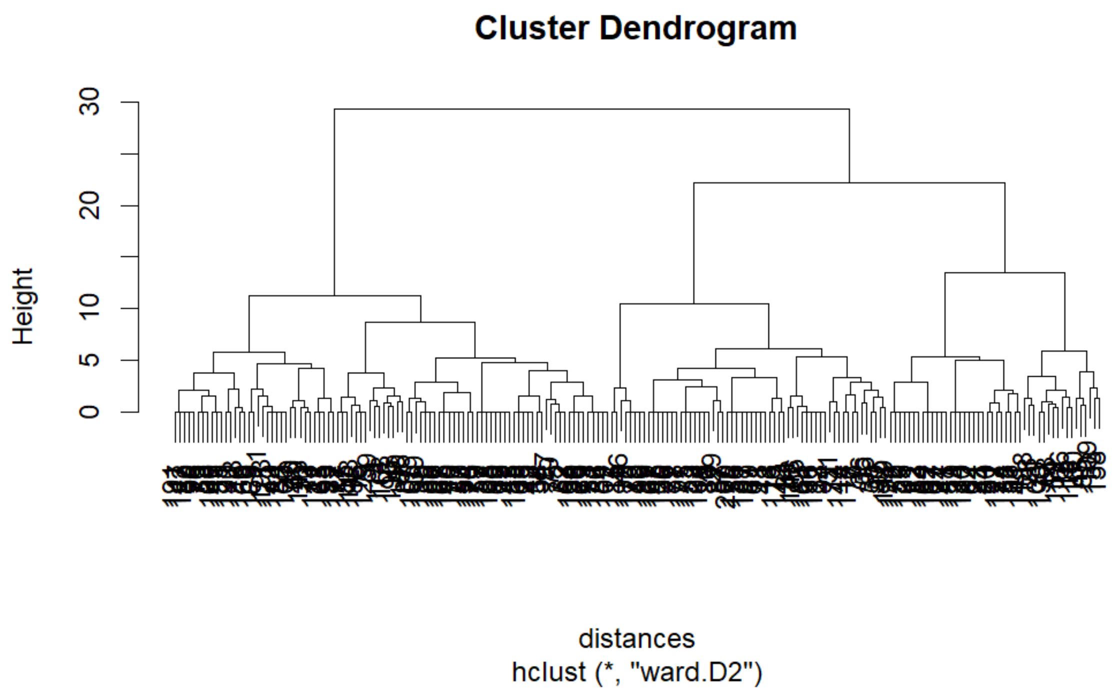

Figure 1 Dendrogram without cluster highlights

Discussion:

- Euclidean distances were computed using standardized store attribute scores to assess dissimilarities. The smallest distance of 1.62 occurs between respondents 3 and 4. This indicates high similarity in their shopping preferences. Respondents 1 and 3 show the highest dissimilarity with a score of 4.07. This suggests contrasting behavior patterns.
- set.seed(123) was applied for reproducibility. This ensures the same labeling of the clusters.
- Hierarchical clustering with the Ward.D2 method was used. This minimizes within-cluster variance and enhances interpretability.
- The dendrogram shows a notable vertical separation between height 20-25. This indicates a natural breakpoint for data segmentation.

Task 8-9: 3-Cluster Solution, Dendrogram Highlighting and Cluster Membership Count

Code:

```
# Task 8:

rect.hclust(hc, k = 3, border = "red")

# Task 9:

hc3 <- cutree(hc, k = 3)

table(hc3)
```

Useful Output:

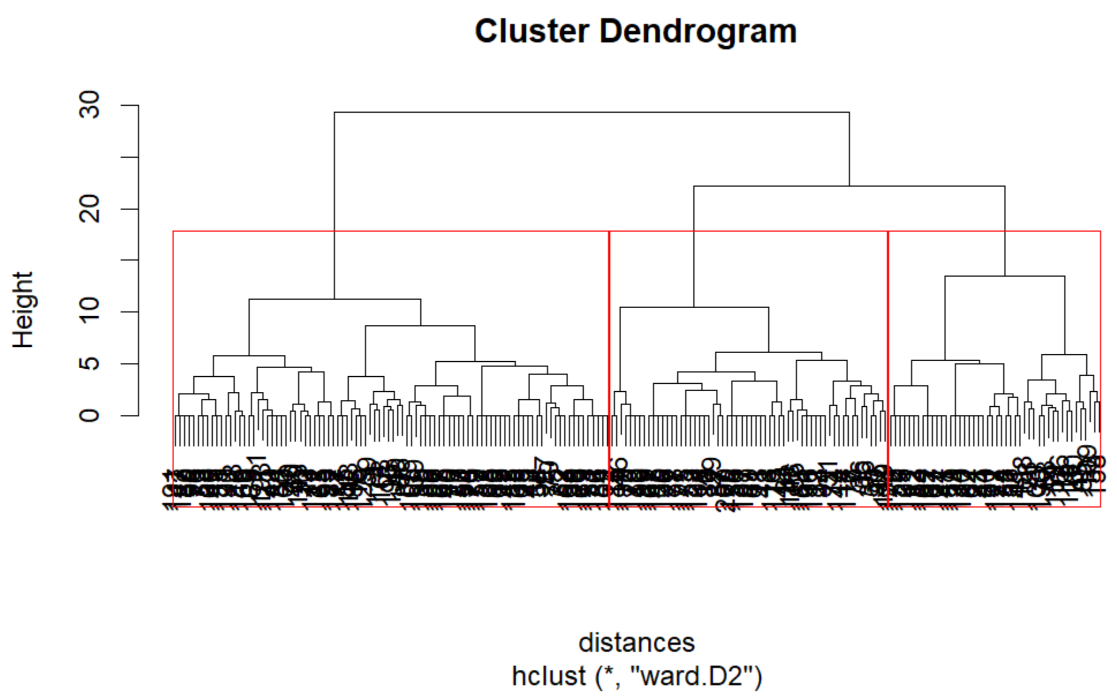

Figure 2 3-cluster Dendrogram with highlights

Table 5 Hierarchical 3-Cluster membership

Cluster	Members
1	94
2	60
3	46

Discussion:

- The dendrogram was successfully segmented into 3 clusters. It is visually confirmed via rect.hclust() and analytically validated using cutree().
- Cluster sizes are as follows:
    1. **Cluster 1:** 94 respondents
    2. **Cluster 2:** 60 respondents
    3. **Cluster 3:** 46 respondents

Task 10: K-Means Clustering (3 Clusters) and Membership Counts

Code:

```
# Task 10:

set.seed(123)

kmeans_3 <- kmeans(store_attributes_norm,
        centers = 3,
        iter.max = 1000,
        nstart = 100)

kmeans_3$size
```

Useful Output:

Table 6 k-means 3-cluster membership

Cluster	Members
1	60
2	46
3	94

Discussion:

- K-means clustering shows similar cluster sizes to hierarchical clustering.
- Cluster sizes are as follows:
    4. **Cluster 1**: 94 respondents
    5. **Cluster 2**: 60 respondents
    6. **Cluster 3**: 46 respondents
- set.seed(123) ensured result reproducibility.
- The validity and reliability of the segmentation is enhanced by the alignment between k-means and hierarchical outcomes.

Task 11: 4-Cluster Solution and Membership Counts

Code:

```
# Task 11:

hc4 <- cutree(hc, k = 4)

plot(hc, labels = FALSE, main = "Dendrogram with 4 Clusters")

rect.hclust(hc, k = 4, border = "red")

hc4 <- cutree(hc, k = 4)
```

table(hc4)

Useful Output:

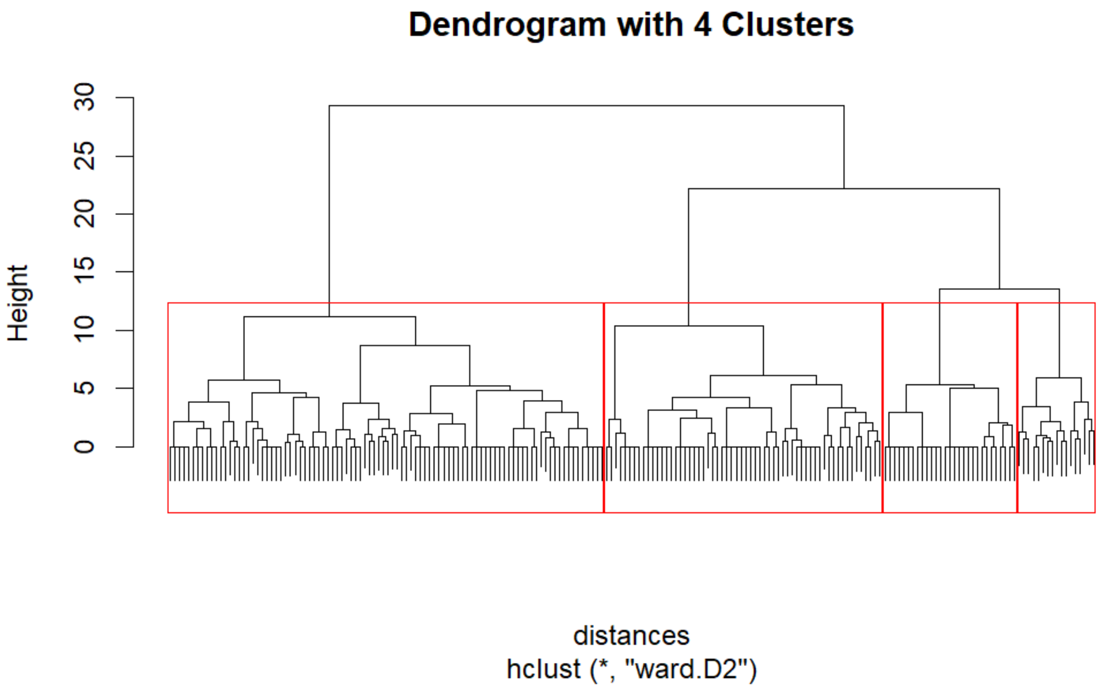

Figure 3 4-cluster Dendrogram

Table 7 Hierarchical 4-cluster membership

Cluster	Members
1	94
2	60
3	17
4	29

Discussion:

- The dendrogram was successfully segmented into 4 clusters. It is visually confirmed via rect.hclust() and analytically validated using cutree().
- The new cluster distribution is:
  1. **Cluster 1:** 94 respondents
  2. **Cluster 2:** 60 respondents

3. **Cluster 3**: 17 respondents
4. **Cluster 4**: 29 respondents

- A smaller, more niche segment (Cluster 3) is introduced by this segmentation while maintaining good separation across the other groups.

Task 12: K-Means Clustering (4 Clusters) and Membership Counts

Code:

```
# Task 12:

set.seed(123)

kmeans_4 <- kmeans(store_attributes_norm,
          centers = 4,
          iter.max = 1000,
          nstart = 100)

table(kmeans_4$cluster)
```

Useful Output:

Table 8 K-means 4-cluster membership

Cluster	Members
1	17
2	60
3	29
4	94

Discussion:

- The 4-cluster k-means solution yielded similar results to a 4-cluster hierarchical solution.
- The new cluster distribution is:
    1. **Cluster 1**: 17 respondents
    2. **Cluster 2**: 60 respondents
    3. **Cluster 3**: 29 respondents
    4. **Cluster 4**: 94 respondents
- set.seed(123) ensures replicability, nstart = 100 avoids convergence on local optima and iter.max = 1000 sets the maximum number of iterations the k-means algorithm can perform for a single run.

Task 13: Cluster Solution Comparison and Validation (NbClust & ward.D2)

Code:

```
# Task 13:

library(NbClust)

library(car)

vif(lm(return_policy ~ ., data = store_attributes_norm))

set.seed(1990)

NbClust(data = store_attributes_norm[, 1:6], min.nc = 3,  max.nc = 15, index = "all", method = "ward.D2") $Best.ncBest
```

Useful Output:

Table 9 Multicollinearity Table (VIF Test)

	variety_of_choice (DV)	**electronics (DV)**	**furniture (DV)**	**quality_of_service (DV)**	**low_prices (DV)**
return_policy (IDV)	2.03	1.30	2.30	1.75	1.65

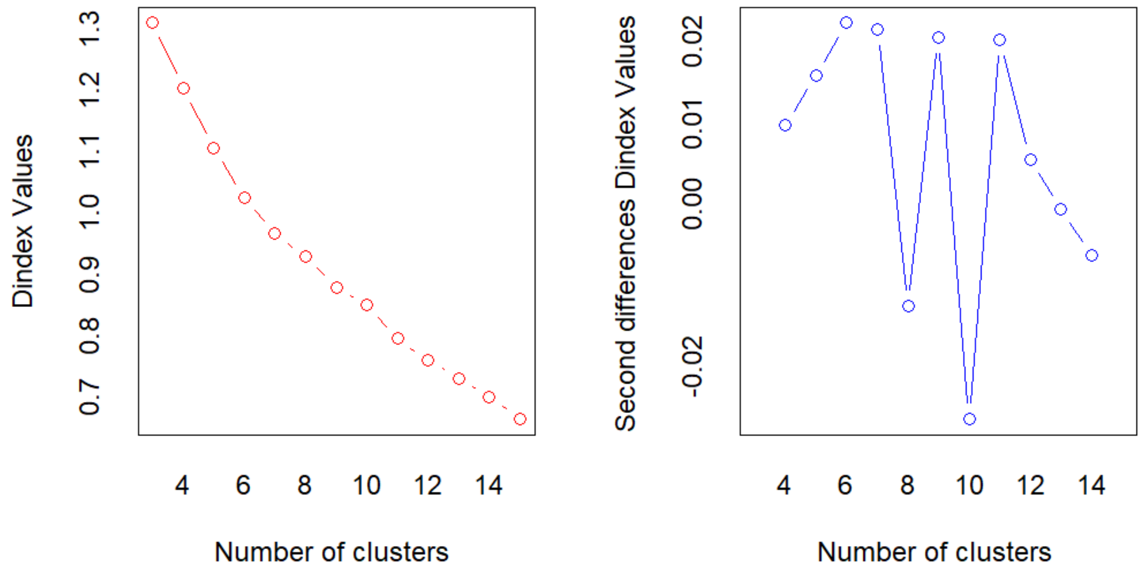

Figure 4 ward.D2 method to determine the best number of clusters

Among all indices:

5 proposed 3 as the best number of clusters

8 proposed 4 as the best number of clusters

2 proposed 5 as the best number of clusters

1 proposed 6 as the best number of clusters

4 proposed 7 as the best number of clusters

1 proposed 10 as the best number of clusters

2 proposed 15 as the best number of clusters

Conclusion: According to the majority rule, the best number of clusters is 4

Discussion:

- For checking multicollinearity among variables, the car package is used to calculate the Variance Inflation Factors (VIF) using vif(), keeping return_policy as the independent variable and all other store attributes as dependent variables. It is found that multicollinearity is not a concern as all VIF values are below the conservative threshold of 5 according to Kim (2019). This confirms the absence of redundant linear relationships among predictors.
- set.seed(123) was applied for reproducibility. NbClust was used with Ward.D2 method to minimize within-cluster variance and determine the best number of clusters.
- NbClust majority rule indicates 4 is the optimal number of clusters which is supported by 8 indices.

Section 2: Segment Description

Task 14-15: Cluster Means, Segment Profile Plot and Segment Description

Code:

```
# Task 14:


library(flexclust)


hc4 <- cutree(hc, k = 4)
```

```
store_attributes_norm %>%

mutate(cluster = factor(hc4)) %>%

group_by(cluster) %>%

mutate(n = n()) %>%

summarise_all(~ mean(.x)) %>%

mutate(prop = n / sum(n)) %>%

print(width = Inf)


hc4_flex <- as.kcca(hc, store_attributes_norm, k = 4)


barchart(hc4_flex)


# Task 15:


retailer %>%

 mutate(cluster = factor(hc4)) %>%

 group_by(cluster) %>%

 summarise(

  mean_income = mean(income),

  mean_age = mean(age)

 )
```

Useful output:

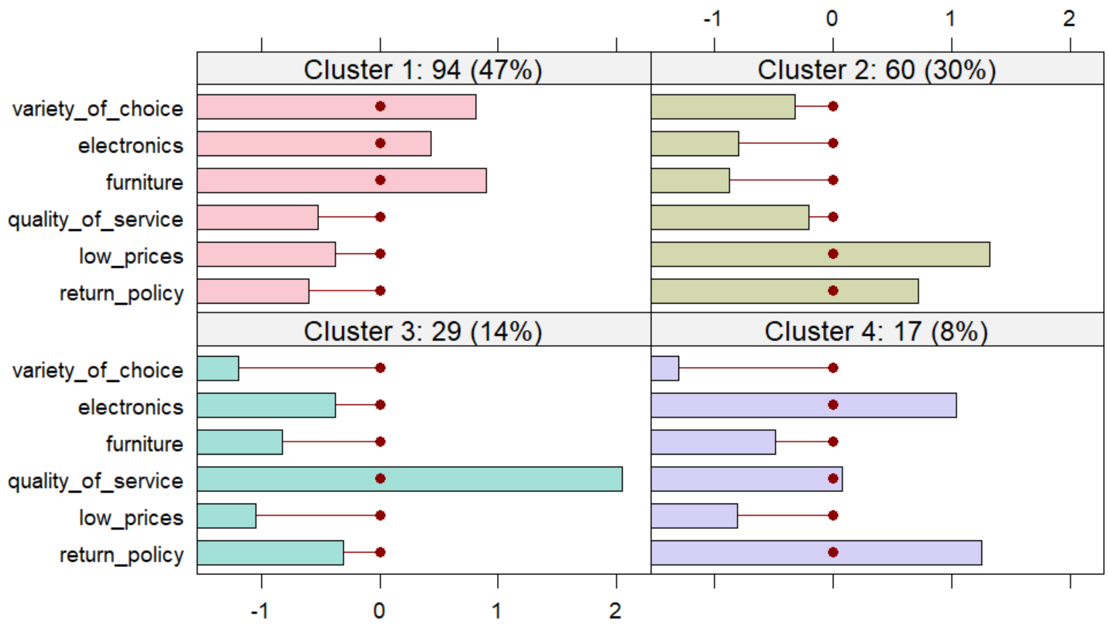

Figure 5 Segment profile plot

Table 10 A tibble: 3 × 9, displaying the means for each store attribute variable of the Normalised data

Cluster	variety_of_choice	electronics	furniture	quality_of_service	low_prices	return_policy	n	proportion
1	0.81	0.44	0.90	-0.52	-0.38	-0.60	94	47%
2	-0.32	-0.80	-0.87	-0.20	1.33	0.72	60	30%
3	-1.30	1.04	-0.49	0.08	-0.80	1.26	17	8.5%
4	-1.20	-0.37	-0.83	2.05	-1.05	-0.31	29	14.5%

Table 11 Mean income & age per cluster

Cluster	mean_income	mean_age
1	31.1	31.6
2	19.0	25.5
3	45.6	38.6
4	54.8	46.2

Discussion:

- flexclust was used for creating the segment profile plot.

Segment 1: 94 respondents (47%)

- **Attitudes and behaviours:** Respondents in this cluster score highest on furniture with 0.90, electronics with 0.44 and variety of choice with 0.81. This shows a clear preference for wide selections and quality tangible goods. They are moderate in quality of service with a score of -0.52 and have low interest in low prices with -0.38.
- **Demographics:** They have an average income of $31.1k and age of 31.6 years. This segment appears to be young professionals with stable incomes who seek well-furnished environments with modest price sensitivity.
- **Name:** Value-Conscious Traditionalists

Segment 2: 60 respondents (30%)

- **Attitudes and behaviours:** This group values low prices with a score of 1.33 and return policy with 0.72, with low preference across product attributes. Their attitude prioritizes price fairness and purchase security.
- **Demographics:** This group have the lowest average income of $19k and the youngest age of 25.5 years. They show cost sensitivity and appear to be students or early-career consumers.
- **Name:** Budget-Focused Pragmatists

Segment 3: 17 respondents (8.5%)

- **Attitudes and behaviours:** Scoring very high on electronics with 1.04 and return policy with 1.26 and above average on quality of service with 0.08. This small but distinct group seeks innovation and post-purchase reliability.
- **Demographics:** This group has an average income of $45.6k and an age of 38.6 years. They appear to be tech-savvy, mid-career customers open to experimenting with new services and digital touchpoints.
- **Name:** Service-Driven Tech Explorers

Segment 4: 29 respondents (14.5%)

- **Attitudes and behaviours:** This group places the highest emphasis on quality of service with a score of 2.05 and below average on all other attributes. Their attitude prioritizes exceptional service over tangible offerings or pricing.
- **Demographics:** This group has the highest income of $54.8k and oldest average age of 46.2 years. These are seasoned customers ideal for loyalty programs and premium experiences.
- **Name:** Experience-Oriented Loyalists

Section 3: Target Segment Evaluation and Strategic Recommendation

The effectiveness of a firm's marketing strategy depends on its ability to align internal capabilities with external market opportunities (Vorhies et al., 2009). Chestnut Ridge needs to ensure that its internal capabilities are effectively matched with strategic priorities by aligning its strengths and weaknesses with the the specific customer segments it aims to serve. Chestnut Ridge can evaluate which customer segments offer the most potential by assessing segment attractiveness alongside the company's business strengths with the help of the GE matrix. It enables informed recommendations on where to invest, manage or divest after a structured comparison of each segment (Amatulli et al., 2011).

Strategic Evaluation of Customer Segments

Value-Conscious Traditionalists

This segment is characterized by strong preferences for tangible product attributes such as furniture, variety of choice and electronics. Being the largest segment with moderate service expectations and price sensitivity makes them the best fit out of the four customer segments for Chestnut Ridge's existing strengths of a broad product range, national distribution and dependable physical retail infrastructure (Giammattei, 2024).

- **Business Strength**: Aligns well with product and operational core – Strong

- **Segment Attractiveness**: Large, growing and accessible – High
- **Strategic Priority**: Invest & Grow / Loyalty Building

Budget-Focused Pragmatists

This segment indicates a clear value-seeking behaviour by ranking the highest for low prices and return policy. However, this segment suggests transactional rather than relational loyalty as their preferences for other attributes are notably low. They are students or early-career shoppers with low average income and young age resulting in limited customer lifetime value (Bhattacharyya & Sarma, 2025). Over-prioritizing this segment may strain margins and dilute brand positioning in the long run although Chestnut Ridge can serve them through their existing infrastructure.

- **Business Strength**: Cost leadership possible at scale – Medium
- **Segment Attractiveness**: Price sensitive and low loyalty potential – Medium
- **Strategic Priority**: Manage Selectively / No long-term investments

Service-Driven Tech Explorers

This niche group shows moderate interest in quality of service but values electronics and return policy. They are mid-career, tech-savvy consumers likely to respond to digital touchpoints, personalised marketing and seamless service experiences (Darvidou, 2024). However, this is a resource-intensive segment to serve without operational upgrades as Chestnut Ridge currently lacks digital sophistication, omnichannel integration or CX tools.

- **Business Strength**: Current infrastructure underdeveloped for this segment – Weak
- **Segment Attractiveness**: Future-focused, affluent and digital – High
- **Strategic Priority**: Manage Selectively / Niche Development

Experience-Oriented Loyalists

With the highest income and oldest average age, this segment prioritizes quality of service. These customers value emotional experience and brand loyalty over product features or price making them a premium market opportunity (Ozuem et al., 2021). However, this segment is desirable but organisationally misaligned in the short term as Chestnut Ridge's heritage in traditional retailing and product diversity has not yet translated into personalised experiences, loyalty programmes or CRM systems.

- **Business Strength**: Capabilities for premium CX not yet in place – Weak.
- **Segment Attractiveness**: Affluent, loyal and high-margin – High.
- **Strategic Priority**: Manage Selectively / Long-Term Investment

GE Matrix

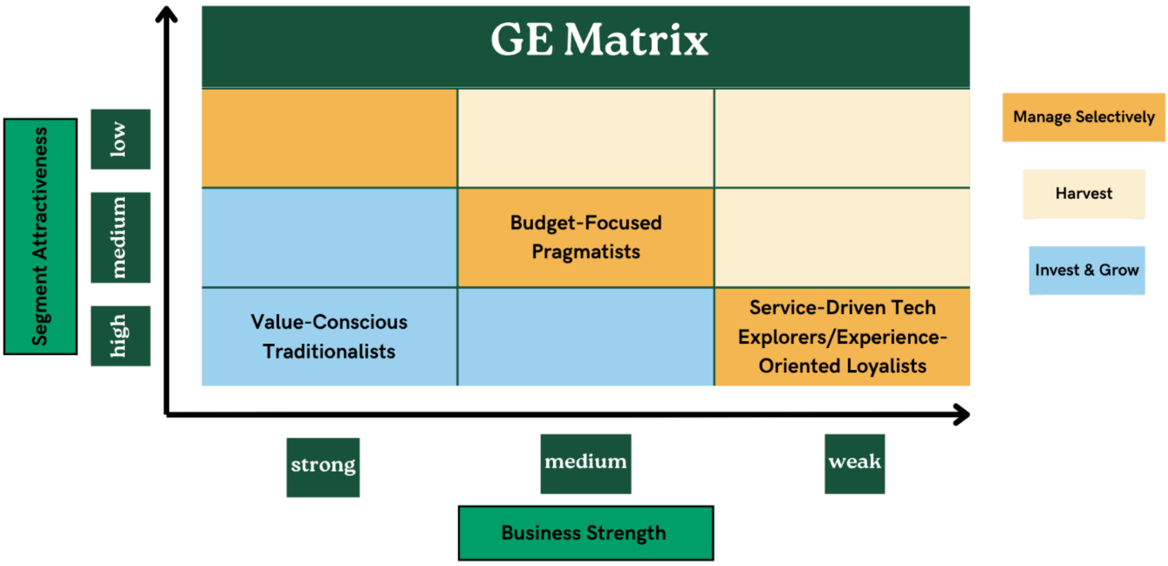

Figure 6 Mckinsey GE Matrix

Table 12 Summary of Analysis

Segment	Business Strength	Segment Attractiveness	Strategic Priority
Value-Conscious Traditionalists	Strong	High	Invest & Grow / Loyalty Building
Budget-Focused Pragmatists	Medium	Medium	Manage Selectively / No long-term investments

Service-Driven Tech Explorers	Weak	High	Manage Selectively / Niche Development
Experience-Oriented Loyalists	Weak	High	Manage Selectively / Long-Term Investment

Strategic Recommendations

Recommendation 1

The most strategically aligned segment to target immediately is the Value-Conscious Traditionalists based on the GE Matrix evaluation and Chestnut Ridge's internal capabilities. Segment size, accessibility, alignment with the company's logistical and product diversification capabilities and mid-level pricing tolerance make it the segment with the strongest fit across all dimensions. Furthermore, targeted in-store experiences and bundling strategies leveraging Chestnut Ridge's historical expertise can increase their loyalty (Grewal et al., 2008). A successful example of a retailer that retains mid-market consumers by offering value bundles, quality store brands and curated in-store promotions is Target in the U.S. (PYMNTS, 2024). These are strategies that Chestnut Ridge can replicate.

Recommendation 2

Budget-Focused Pragmatists should be retained through standard promotions, seasonal offers and competitive return policies (Karlsson et al., 2023). However, they should not be prioritized for long-term investment due to their high price sensitivity and low switching costs. Walmart excels in this space through its Everyday Low Price (EDLP) strategy for attracting price sensitive customers (Choo, 2024). Chestnut Ridge can similarly sustain traffic and sales through tactical offers.

Recommendation 3

Service-Driven Tech Explorers and Experience-Oriented Loyalists should be developed incrementally although they are highly attractive in terms of income and behavioural loyalty. These segments can be effectively targeted only after strategic investments in omnichannel capabilities, customer data platforms and loyalty ecosystems (Darvidou, 2024). For instance, Nordstrom’s Nordy Club loyalty program and high-touch customer service are central to their premium customer experience model (Hure, 2024). A membership-based subscription strategy tailored specifically to Segment 4's needs is a smart approach for Chestnut Ridge.

Conclusion

This report used a data-driven approach to explore Chestnut Ridge’s customer base, combining hierarchical and k-means clustering techniques to uncover meaningful segments. Four distinct customer groups, each with its own set of preferences, income levels and shopping behaviours were revealed through the analysis. These groups told a clear story about what different types of customers truly value, from affordability and variety to service quality and product innovation hence, these differences were not just statistical.

The Value-Conscious Traditionalists stood out as the best fit for Chestnut Ridge’s current strengths, including its wide product range and multichannel reach through the application of the GE Matrix. The Budget-Focused Pragmatists are less ideal for long-term loyalty although they offer short-term volume potential. The more affluent Tech Explorers and Experience-Oriented Loyalists would require deeper investment in digital tools and customer experience strategies as they show great promise.

Ultimately, this segmentation gives Chestnut Ridge a focused, informed and built-to-last path, moving forward in a fast-changing retail world.

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